

READING, PA, USA

WMO: 725103

Lat: 40.373N

Lon: 75.959W

Elev: 344

StdP: 14.51

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14712

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	13.6	-5.1	4.2	12.1	-0.4	5.4	16.5	27.0	30.7	24.6	29.2	9.1	290	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	92.3	74.6	89.6	73.6	87.0	72.2	77.5	87.5	76.0	84.9	74.7	82.9	8.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	131.2	81.9	73.2	124.9	80.2	72.2	120.6	79.3	41.1	87.5	39.5	85.1	38.3	82.8	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.2	19.5	17.4	DB	5.1	97.4	5.0	5.8	1.5	101.5	-1.4	104.9	-4.2	108.2	-7.8	112.4
			WB	3.8	80.2	4.9	2.2	0.3	81.8	-2.6	83.1	-5.3	84.3	-8.9	85.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	30.7	32.8	41.3	52.9	62.7	71.3	76.0	74.0	67.2	55.4	44.6	35.7
	DBStd	17.70	10.32	9.32	9.32	8.48	7.72	6.15	5.18	5.11	6.71	7.94	8.16	8.66
	HDD50	2139	602	486	300	62	2	0	0	0	0	33	201	453
	HDD65	5125	1063	902	735	376	141	17	0	3	53	314	611	910
	CDD50	3538	4	3	31	147	395	639	806	745	517	202	40	9
	CDD65	1047	0	0	1	12	69	206	341	282	119	17	0	0
	CDH74	9356	0	1	13	156	691	1830	3271	2367	909	116	2	0
	CDH80	3289	0	0	1	41	221	609	1329	809	261	18	0	0
Wind	WSAvg	6.4	7.6	7.8	8.5	7.8	6.1	5.6	5.1	4.4	4.9	5.7	6.6	7.0
Precipitation	PrecAvg	42.7	3.2	2.6	3.3	3.5	3.9	4.1	4.3	3.5	4.1	3.1	3.5	3.3
	PrecMax	60.5	9.8	5.6	6.5	7.9	11.7	15.1	11.7	7.8	11.2	10.7	8.0	8.5
	PrecMin	27.7	0.5	0.6	0.6	0.8	0.9	0.8	0.5	0.2	0.3	0.2	0.5	0.8
	PrecStd	7.4	2.0	1.2	1.4	1.8	2.0	2.6	2.3	2.0	2.6	2.2	1.8	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	64.5	76.2	85.0	90.1	93.4	98.4	94.5	91.0	82.3	71.4	65.5
		MCWB	60.4	56.1	60.6	65.3	72.0	74.9	76.0	74.8	73.4	69.1	61.7	59.9
	2%	DB	56.7	57.6	67.5	79.1	86.1	89.7	93.2	90.3	86.3	77.4	66.4	58.8
		MCWB	52.7	48.3	55.2	62.4	69.9	74.0	75.2	73.9	71.0	67.1	58.9	54.6
	5%	DB	49.4	51.6	61.9	73.2	81.8	86.5	90.2	87.2	82.4	73.1	62.9	53.6
		MCWB	45.1	44.1	52.0	59.9	67.6	72.0	74.3	72.6	69.5	64.5	56.0	49.3
	10%	DB	44.8	46.1	56.6	68.5	77.3	83.6	87.3	84.3	79.4	69.6	58.2	48.4
		MCWB	40.8	40.4	48.0	56.4	64.8	70.1	72.9	71.4	67.7	61.7	51.9	43.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.0	58.9	62.1	68.9	74.8	77.3	80.8	79.2	76.7	73.2	64.4	60.3
		MCDB	64.9	63.2	71.7	80.2	86.1	88.3	91.3	89.9	85.2	77.3	68.3	63.6
	2%	WB	53.6	50.8	57.2	65.4	72.1	75.6	78.0	76.9	74.3	69.6	60.9	55.1
		MCDB	56.3	56.2	64.7	75.0	82.6	85.6	88.9	85.6	81.1	75.1	64.9	58.6
	5%	WB	46.0	45.1	53.0	62.0	69.7	74.2	76.4	75.2	72.5	66.4	56.9	50.0
		MCDB	48.8	49.8	59.8	70.7	78.6	83.3	85.8	82.8	78.8	71.9	61.5	52.7
	10%	WB	41.3	41.0	48.4	58.6	67.2	72.6	74.9	73.8	70.7	62.8	53.3	44.5
		MCDB	44.5	46.4	55.5	66.9	74.8	80.7	83.4	80.7	76.7	67.6	57.3	47.8
Mean Daily Temperature Range	5% DB	MDBR	14.4	16.0	17.8	20.4	20.0	19.1	19.4	18.5	19.1	18.8	17.1	13.9
		MCDBR	21.3	23.2	26.0	27.7	25.4	22.5	22.7	21.2	22.8	23.3	23.0	19.9
		MCWBR	17.9	16.9	17.0	15.2	12.0	9.5	8.3	8.4	10.2	13.1	16.1	16.7
	5% WB	MCDBR	20.0	20.8	23.7	23.5	22.1	19.6	19.2	17.7	17.6	18.8	18.8	17.7
		MCWBR	18.0	16.9	17.7	15.0	11.7	9.6	8.4	8.2	9.5	12.5	15.7	16.6
Clear-Sky Solar Irradiance	taub	0.309	0.327	0.360	0.393	0.433	0.498	0.498	0.497	0.411	0.364	0.326	0.307	
	taud	2.436	2.359	2.354	2.287	2.209	2.064	2.079	2.069	2.280	2.421	2.492	2.485	
	Ebn at Noon	270	280	283	279	269	251	250	246	262	264	262	262	
	Edh at Noon	25	31	36	41	45	52	51	50	38	30	24	23	
All-Sky Solar Radiation	RadAvg	578	864	1150	1492	1673	1836	1876	1629	1341	945	662	491	
	RadStd	50	103	88	141	199	153	96	98	172	104	62	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-216	+310	N/A
Regional (70 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	+133

Nomenclature: See separate page